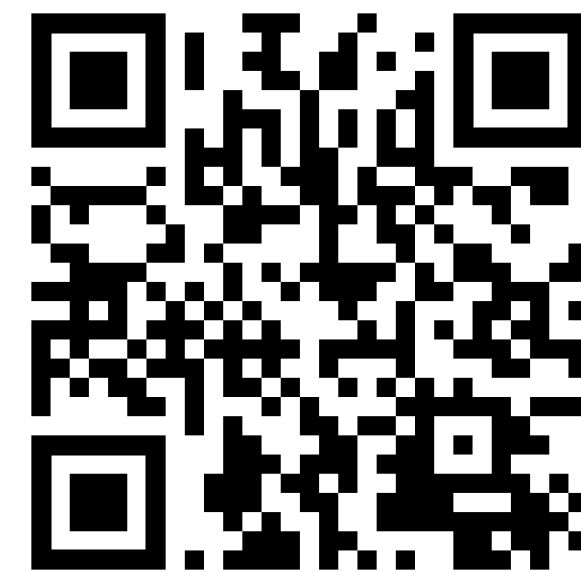


(Most) Russian palatalisation contrasts use tongue root

Tehya Banach, Denis Tokmashev, Jonathan Washington



Palatalisation contrast

- Russian consonants (Cs) occur in C^j and C^ɣ pairs in **overlapping distribution**.
[ɯuk] ‘hatch (cover)’ [ɮuk] ‘onion’
- Previous studies:
 - C^j has raised, **advanced** tongue body (TB)
 - C^ɣ has **retracted** tongue root (TR)
- Ultrasound studies of **other languages**:
 - TR advancement / TB backing primary correlate of C^j/C^ɣ
 - TR position major correlate of vowel (V) backness, even for adjacent Cs

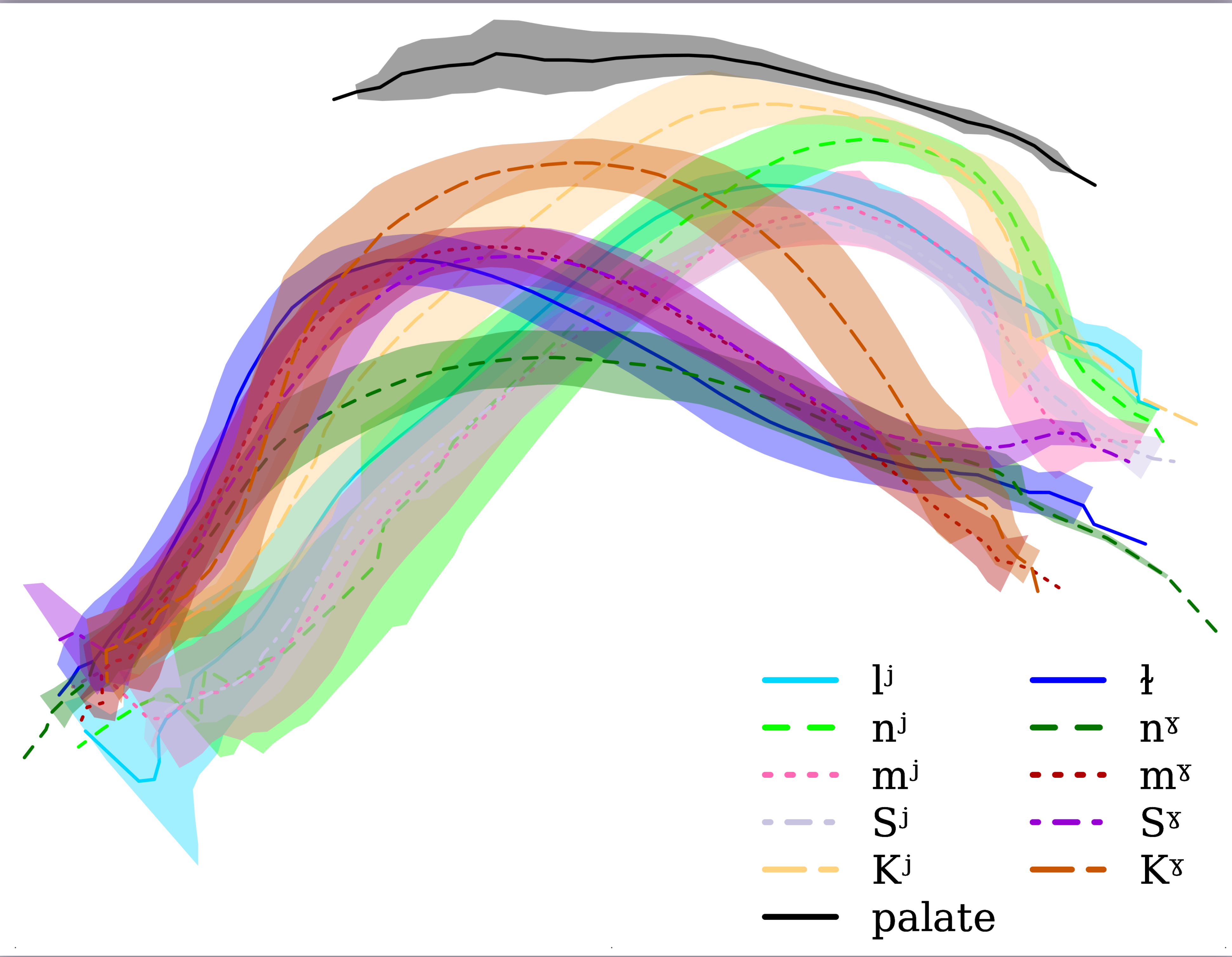
Methodology

- Ultrasound **recordings**:
 - recorded by author 2
 - stimuli in carrier sentence (rep. 3x):
Слово X отличное. ‘The word X is splendid.’
 - interspersed ‘run tongue along palate’
- Annotated for adjacency to **front** (i,e) and **back** (a,o,u) Vs
- Middle frames **traced** and **plotted** to depict mean and standard deviation

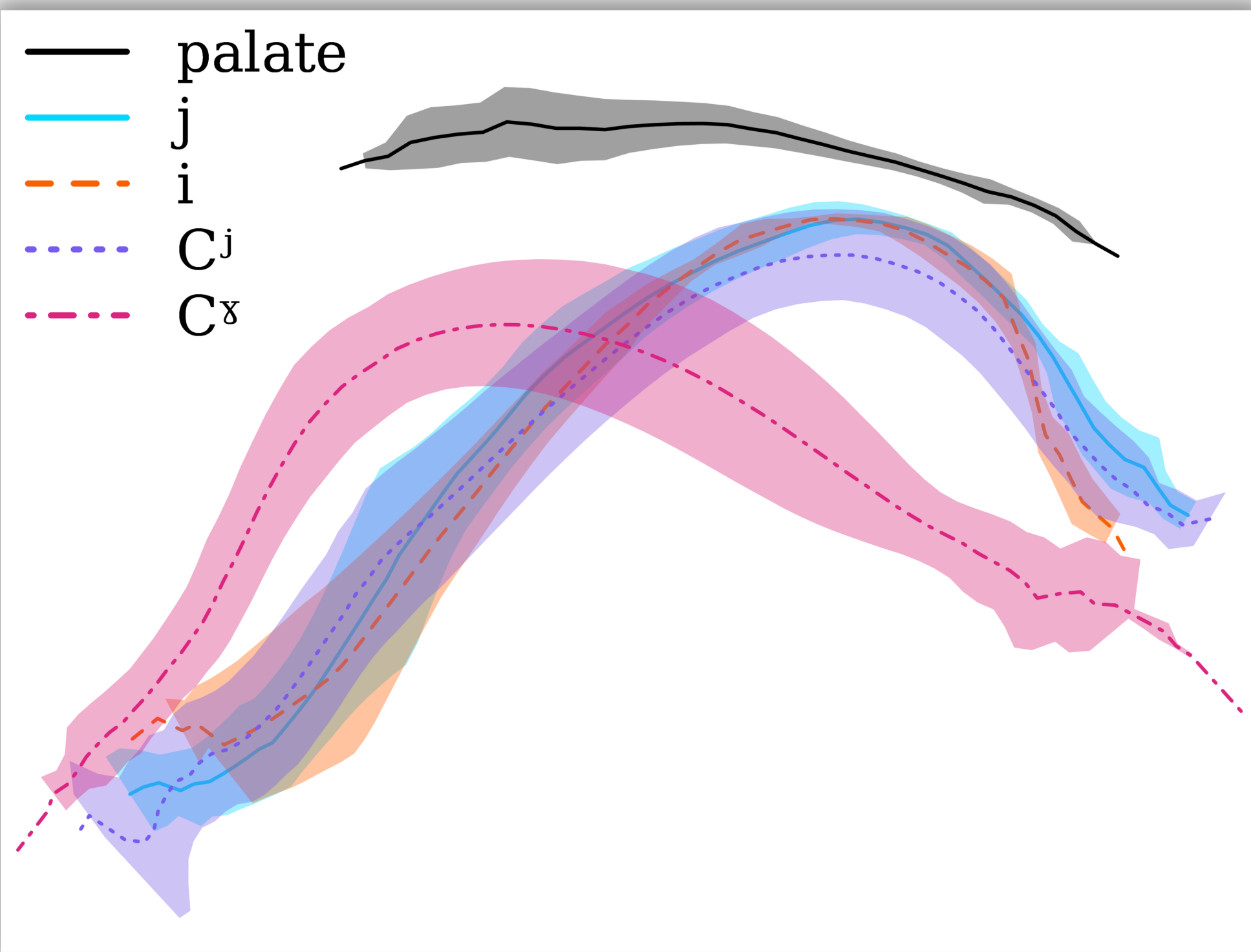
Findings

- For C^j, **TB is raised** and **TR is advanced** in the pharynx.
- For C^ɣ, **TB is low** and **TR is retracted**.
- /K^j/~ /K^ɣ/ and /n^j/~ /n^ɣ/ are **exceptions**:
 - /C^j/: very raised TB (true palatal [ɲ]?)
 - /n^ɣ/: lowered tongue dorsum (plain [n]?)
 - /K^ɣ/: medial TR position
- Adjacent **vowel backness** has mostly **negligible impact** on tongue position

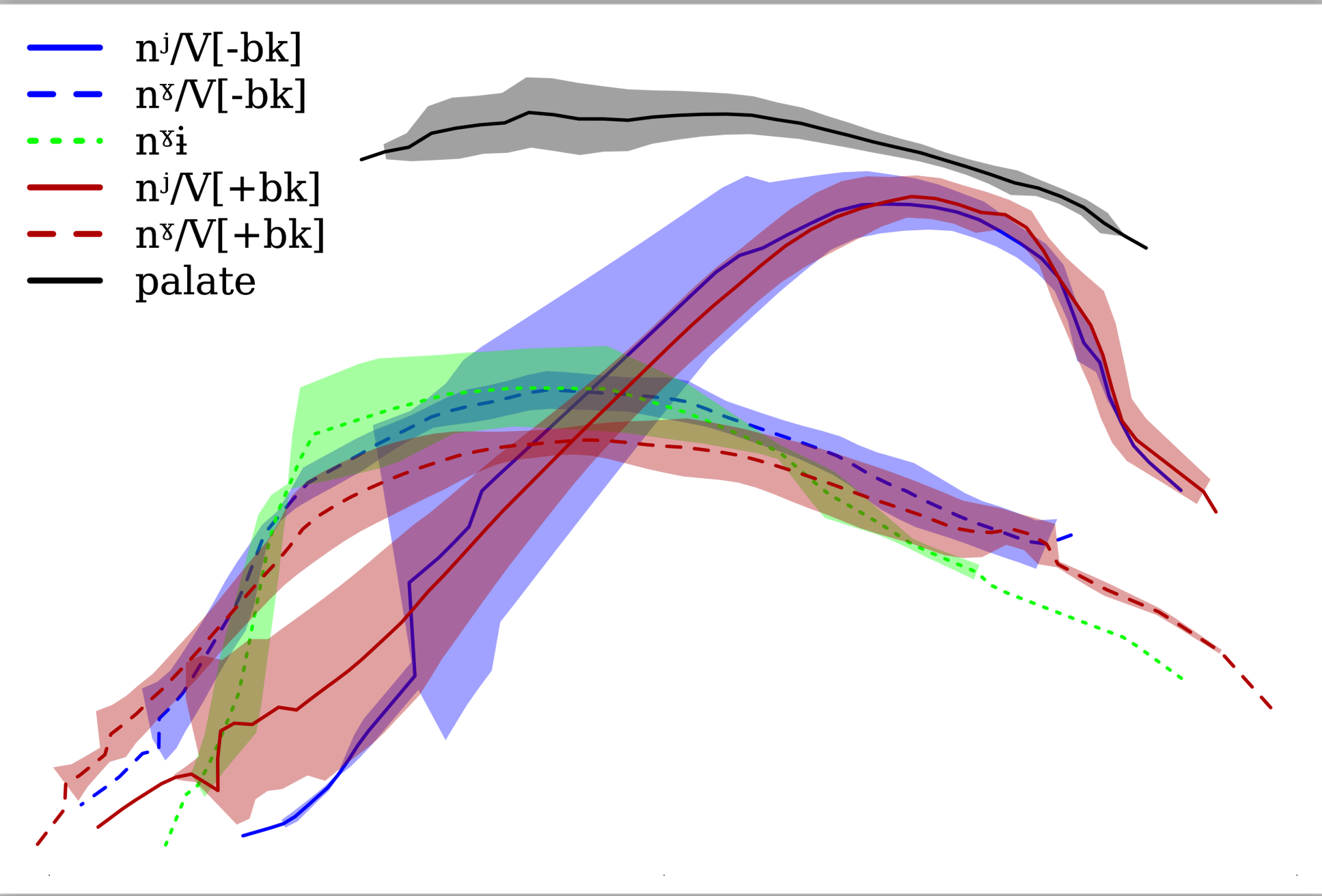
The **palatalised/velarised** distinction in Russian is implemented through both **tongue body** and **tongue root** position.



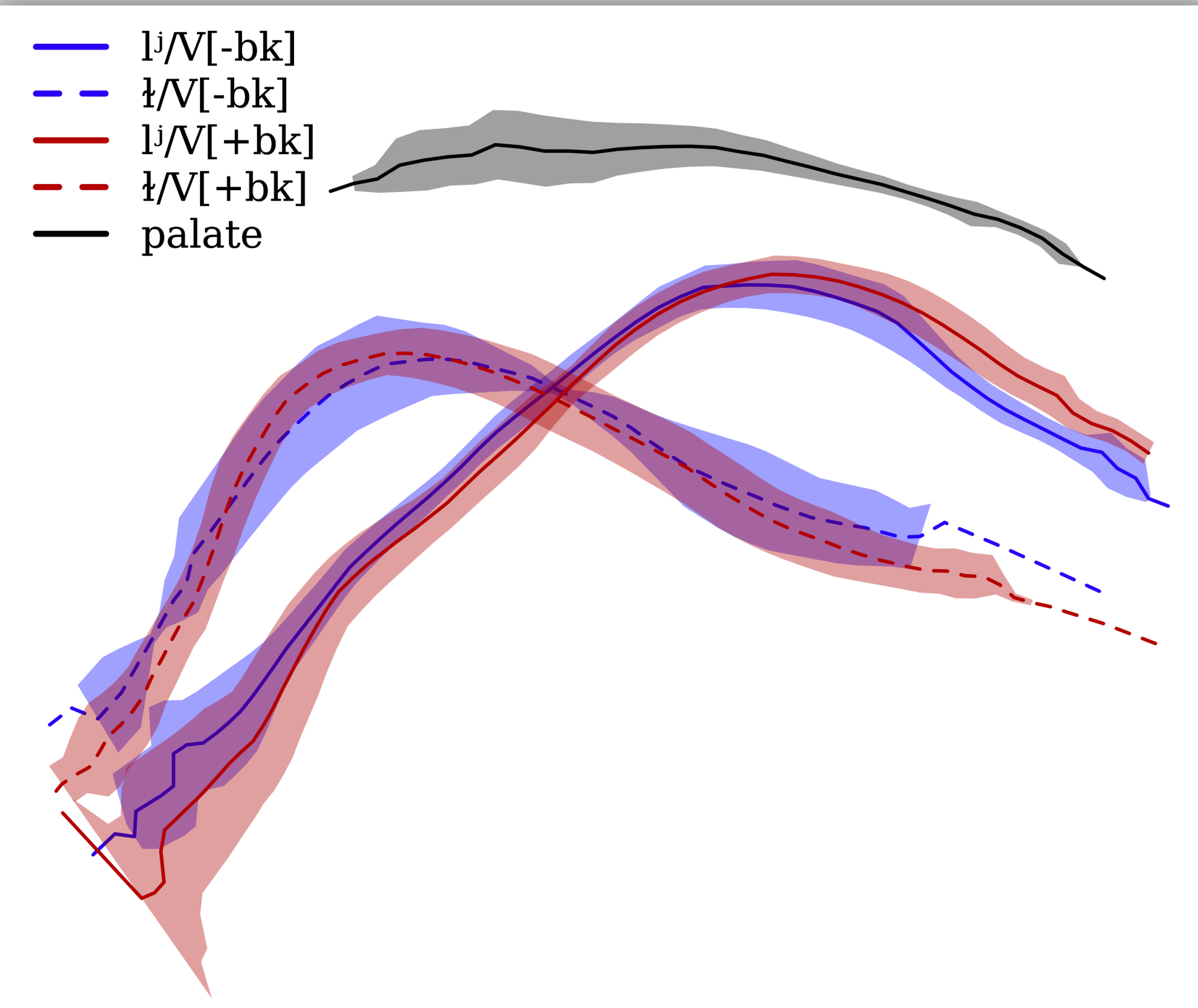
All C^js (n=64) and C^ɣs (93) examined, S={s,z}, K={k,g} [tongue tip right →]



All C^js and C^ɣs combined, with /i/ (5), /j/ (10), and palate (52)



/n^j/ & /n^ɣ/ in different vocalic contexts:
n^j/V[-bk] (5), n^ɣ/V[-bk] (3), n^j/i (2), n^j/V[+bk] (7), n^ɣ/V[+bk] (8)



/l^j/ and /l^ɣ/ in different vocalic contexts:
l^j/V[-bk] (6), l^ɣ/V[-bk] (3), l^j/V[+bk] (9), l^ɣ/V[+bk] (20)

References

Outkin & Washington. 2024. ultrapolaRplot: A free/open-source R library for plotting tongue traces. Murphy et al. 2020. UltraTrace: A free/open-source cross-platform tool for manual annotation of ultrasound tongue imaging data. Cavar et al. 2020. Palatalization in coronal consonants of Polish: a three-/four dimensional ultrasound study. Cavar & Lulich. 2021. Variation in the articulation of Russian stressed vowels and the mechanics of palatalization in consonants. Barry. 1992. Palatalisation, assimilation, and gestural weakening. Padgett. 2001. Contrast dispersion and Russian palatalization. Bennett et al. 2018. An ultrasound study of Connemara Irish palatalization and velarization.

Acknowledgements

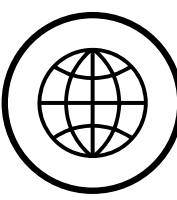
Swarthmore College Academic Division Funding
Emily Gasser
Grant «Языковое и культурное разнообразие Южной Сибири в синхронии и диахронии: взаимодействие языков и культур» / «Linguistic and cultural diversity of Southern Siberia in synchrony and diachrony: interaction of languages and cultures»



National Research
Toms State University



jonathan.washington@swarthmore.edu,
tehyabanach@gmail.com,
kogutei@yandex.ru



phonlab.swarthmore.edu